Contact

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www.linkedin.com/in/alireza-malekmohammadi-57049769 (LinkedIn)

Top Skills

Experimental Research

Research Support

Higher Education

Languages

Persian (Native or Bilingual)

German (B1) (Limited Working)

English (Full Professional)

Arabic (Elementary)

Certifications

REVIEWER CERTIFICATE: Experimental Brain Research

REVIEWER CERTIFICATE

Test Certificate for TUM Deutsch Course from the Technical University of Munich!

REVIEWER CERTIFICATE: Biological Psychology

REVIEWER CERTIFICATE: Experimental Brain Research

Honors-Awards

Third rank among Digital Electronics students, Sharif University of Technology.

Score A+ in the workshop “Design and implementation of digital circuits based on Xilinx FPGAs with Plan Ahead” at Sharif University of Technology.

Score A in the workshop “Effective implementation of digital systems based on Xilinx FPGAs theoretically and practically” at professional level at Sharif University of Technology.

Score A in the workshop “Effective implementation of digital systems

Alireza Malekmohammadi

Postdoctoral Researcher in Brain Networks @ MPI CBS

Leipzig, Saxony, Germany

Summary

Alireza Malekmohammadi is a postdoctoral researcher at the Max Planck Institute for Human Cognitive and Brain Sciences in Leipzig.Focus: Investigation of the working mechanisms underlying the effects of transcranial magnetic stimulation (TMS) of the human motor cortex using advanced electrophysiological measurement techniques combined with biophysical modeling.He finished his Ph.D. in Electrical Engineering at the Technical University of Munich (TUM), Department Computer Engineering. He worked on the "Brain Sound Computer Interface" (utilizing deep and machine learning methods to decode sound sequences from the human brain) at the Institute for Cognitive Systems (ICS, https://www.ei.tum.de/ics/people/malekmohammadi-alireza/).He received his bachelor’s degree in “Electrical Engineering” under the supervision of Prof. Mohammad Eshghi from Shahid Beheshti University, Tehran, Iran , in 2013. In 2015, he was awarded his M.Sc. degree in “Digital Electronics Engineering” from Sharif University of Technology in Iran . During his master's, he attended the Brain-Computer Interface (BCI) group, under the supervision of Prof. Mahdi Shabany, developing an efficient design and hardware implementation of machine learning methods (i.e., SVM, LDA, KNN, ANN) for a BCI system based on motor imagery on a Virtex-6 FPGA. From 2017 to 2018, he spent one year as a research assistant at Tübingen University in Germany . He studied the effects of intra-operative monopolar deep brain stimulation of the subthalamic nucleus on cortical physiology in Parkinson’s disease.

Experience

Max Planck Institute for Human Cognitive and Brain Sciences

Postdoctoral Researcher

May 2026 - Present (7 months)

Leipzig, Saxony, Germany

based on Xilinx FPGAs theoretically and practically" at intermediate level at Sharif University of Technology.

Score 100 (out of 100) in the workshop "RTL design and implementation with FPGA" was organized by ICEEP at Tehran University.

Publications

Modulation of theta and gamma oscillations during familiarization with previously unknown music

Listening to familiar music induces continuous inhibition of alpha and low-beta power

An Efficient Hardware Implementation for a Motor Imagery Brain Computer Interface System

Online determination of the STN "sweet spot" based on beta-power measured with directional macro-contacts of the final DBS lead: 2074.

Speech prediction of a listener via EEG-based classification through subject-independent phase dissimilarity model

Technical University of Munich
PhD Researcher And Teaching Associate
January 2020 - April 2026 (6 years 4 months)
Munich, Bavaria, Germany

Practical Course Biosignal Processing and Modeling (Since 2021; four semesters):
--> Theoretical lectures and hands-on practical tutorials on state-of-the-art sensor systems
--> Methods in experiment design
--> Offline and online (with Openvibe) signal processing, i.e., filtering, ICA
--> Introduction for pattern recognition, i.e., CSP, BFCSP, SCSSP, temporal/ spectral features
--> Introduction for feature reduction, i.e., Fisher criterion, Auto Encoder, PCA
--> Introduction for machine learning, i.e., SVM, LDA, KNN, ANN, naive Bayesian

Practical course neural signals (Since 2020; five semesters):
--> Introduction to (multi-channel) recording, especially EEG, for Brain-Computer Interfaces.
--> Event-related potential (ERP)
--> Wavelet transform
--> Statistical Analysis and multiple comparison correction

Literature Seminar, Scientific Debating, Colloquium (Since 2022; three semesters):
--> Session 1: BCI for communication and control
--> Session 2: Decoding neural signals
--> Session 3: Augmentation of human-machine interaction (HMI)
--> Session 4: BCI for rehabilitation

TUM - Institute for Cognitive Systems
Event Coordinator
September 2024 - June 2025 (10 months)
Washington, District of Columbia, United StatesWashington, District of Columbia, United States

Following the success of two Georgetown-TUM Summer Schools in 2023 and 2024, the third summer school was held at Georgetown University in Washington, DC, in 2025. The theme is "Neuroengineering for Restoration to Function", highlighting ways in which neuroengineering approaches can aid in

restoring function after injury or disease-related impairment. I was honored to be one of the main members on behalf of the TUM coordinators.

The goal of neuroengineering is to apply engineering principles and techniques to understand, repair, replace, and enhance the nervous system. The interdisciplinary nature of the field requires a close collaboration between neuroscientists (spanning approaches from cellular to cognitive) and engineers (including biomedical, computer science, electrical, mechanical…), and the goal of the summer school is to bring together students (from advanced undergraduate to graduate), postdocs, and faculty from all these areas.

TUM - Institute for Cognitive Systems

Event Organizer

September 2023 - June 2024 (10 months)

Munich, Bavaria, Germany

The TUM-Georgetown Neuroengineering Summer School is an annual event, jointly organized by the Center of Competence in Neuroengineering at TUM and the Center for Neuroengineering at Georgetown University. The first TUM-Georgetown Neuroengineering Summer School was held in Washington in 2023. The aim of the program is to educate the next generation of human-centered neuroengineers. Neuroengineering is an emerging interdisciplinary field that focuses on translating findings in neuroscience into practical engineering applications to enhance the quality of life for individuals with neurological and sensorimotor disabilities. In turn, neuroscientists are gaining new insights into brain function based on these advanced neuroengineering techniques. To fully harness the potential of neurotechnologies for patients and society as a whole, we need to educate a new generation of neuroengineers who are proficient in the interdisciplinary integration of neuroscience, engineering, and neuroethics.

I am very grateful to have been one of the main organizers on behalf of the TUM members and to have organized the annual event of the TUM-GU Neural Engineering Summer School 2024.

Georgetown University

40 days internship

August 2023 - September 2023 (2 months)

District of Columbia, United States

Studying the music perception in the monkey's brain during passive listening to music.

TUM - Institute for Cognitive Systems
Event Coordinator
March 2023 - June 2023 (4 months)
Washington, District of Columbia, United States

The Center for Neuroengineering at Georgetown University hosted the first Summer Session in Neuroengineering in partnership with the Technical University of Munich from June 19th to June 23rd, 2023, in Washington, DC. The topic for this exciting event is “Neuroengineering: Bridging Neuroscience and Engineering".

I am very grateful to have been one of the main organizers on behalf of the TUM members and to have contributed to the organization of the TUM-GU Neural Engineering Summer School 2023.

Georgetown University
3.5 months internship
March 2020 - June 2020 (4 months)
District of Columbia, United States

He studied the effect familiary of action in the monkey's brain during passive listening to music.

(BMW) Munich Motors
Intelligent Cars on Digital Roads - Frontiers in Machine Intelligence
July 2018 - August 2018 (2 months)
Bayern, Bavaria, Germany

Attending the 6th French-German summer school for PhD candidates and junior scientists

July 29 - August 03, 2018, Raitenhaslach, Bavaria, Germany

Universitätsklinikum Tübingen
Researcher
April 2017 - April 2018 (1 year 1 month)
Greater Tubingen Area

Sharif University of Technology
Teacher Assistant
October 2014 - December 2014 (3 months)
Iran,Tehran

Logic Circuit (LAB) Dr.M.Shabany

Shahid Beheshti University
Teacher Assistant
September 2012 - May 2013 (9 months)
Iran,Tehran

Microelectronic Dr.M.rashtian

Shahid Beheshti University
Teacher Assistant
September 2010 - January 2011 (5 months)
Iran,Tehran

Mathematical 1 Dr. K. Paran

Education

Technical University of Munich
Dr.-Ing (equivalent to PhD), Electrical and Electronics Engineering · (May 2018 - April 2026)

University of Tübingen
Reseach Assistance , Neuroscience · (2017 - 2018)

Sharif University of Technology
Master of Science (MS), Digital Electronics Engineering · (September 2013 - September 2015)

Shahid Beheshti University
Bachelor of Science (BS), Electronics Engineering · (September 2009 - July 2013)

Allameh Tabatabai High School
pre university, Mathematics · (2008 - 2009)